

Math 445/545. Winter 2026. Practice problems for the Midterm 2.

The problems below illustrate various techniques you are supposed to know for the final. Some of the problems below are more difficult than the ones you will see on the midterm.

Problem 1. In the ring $\mathbb{Z}$, find a positive integer a such that $\langle a \rangle = \langle 6 \rangle + \langle 8 \rangle$ (here $+$ stands for the sum of ideals).

Solution: By definition the ideal $\langle 6 \rangle + \langle 8 \rangle = \{a + b \mid a \in \langle 6 \rangle, b \in \langle 8 \rangle\}$. In other words $\langle 6 \rangle + \langle 8 \rangle$ consists of integers which are sums of two summands: one divisible by 6 and another divisible by 8. Clearly all such numbers are even. Also $2 = 6 \cdot (-1) + 8 \cdot 1$; so any even number is contained in $\langle 6 \rangle + \langle 8 \rangle$ (since this is an ideal). Hence $\langle 6 \rangle + \langle 8 \rangle = \langle 2 \rangle$.

Answer: $\langle 6 \rangle + \langle 8 \rangle = \langle 2 \rangle$.

Problem 2. Is the ideal $N = \langle 35 \rangle \subset \mathbb{Z}$ maximal? How many ideals are strictly in between of N and $\mathbb{Z}$?

Solution: the maximal ideals of $\mathbb{Z}$ are of the form $\langle p \rangle$ where p is a prime number; thus $N = \langle 35 \rangle$ is not maximal. If $N = \langle 35 \rangle \subset M = \langle m \rangle$ then $35 \in N$ should be divisible by m ; thus we have just 4 options for m : 1, 5, 7, 35 (all positive divisors of 35). Clearly $N = \langle 35 \rangle$ and $\mathbb{Z} = \langle 1 \rangle$; thus we have exactly 2 ideals strictly in between of N and $\mathbb{Z}$: $\langle 5 \rangle$ and $\langle 7 \rangle$.

Answer: there are exactly 2 ideals strictly in between of N and $\mathbb{Z}$: $\langle 5 \rangle$ and $\langle 7 \rangle$.

Problem 3. Let $N = \langle 2 \rangle \subset \mathbb{Z}$. Prove that $N[x] \subset \mathbb{Z}[x]$ is an ideal of $\mathbb{Z}[x]$. Is the ideal $N[x]$ maximal? Is it prime?

Solution: the ring $N[x]$ consists of all polynomials with even integer coefficients; equivalently $N[x]$ consists of polynomials of the form $2p(x)$ where $p(x) \in \mathbb{Z}[x]$ (thus $N[x]$ is the principal ideal of $\mathbb{Z}[x]$ generated by 2). This is clearly an ideal: it is closed under addition and subtraction since $2p(x) \pm 2q(x) = 2(p(x) \pm q(x))$; it is also closed under multiplication by any other polynomial since $2p(x) \cdot r(x) = 2(p(x)r(x))$.

For any homomorphism $\phi : R \rightarrow S$ of rings we have a homomorphism $\phi_1 : R[x] \rightarrow S[x]$ such that $\phi_1(a_0 + a_1x + \dots + a_nx^n) = \phi(a_0) + \phi(a_1)x + \dots + \phi(a_n)x^n$ (check that this is really a homomorphism!). Let us apply this to the case $R = \mathbb{Z}$ and $S = \mathbb{Z}_2$; we get a homomorphism $\phi_1 : \mathbb{Z}[x] \rightarrow \mathbb{Z}_2[x]$. It is clear that this homomorphism is surjective (since the homomorphism $\phi : \mathbb{Z} \rightarrow \mathbb{Z}_2$ is surjective). The kernel of ϕ_1 is precisely $N[x]$. Thus by the fundamental homomorphism theorem we have $\mathbb{Z}[x]/N[x] \simeq \mathbb{Z}_2[x]$. Thus the quotient $\mathbb{Z}[x]/N[x]$ is not a field (e.g. $x \in \mathbb{Z}_2[x]$ is not invertible); thus $N[x]$ is not maximal ideal. On the other hand $\mathbb{Z}_2[x]$ is an integral domain (since $\mathbb{Z}_2$ is a field, so it is an integral domain). Thus $N[x]$ is a prime ideal.

Answer: the ideal $N[x] \subset \mathbb{Z}[x]$ is prime but not maximal.

Problem 4. How many elements are in $\mathbb{Z}[i]/\langle 3+i \rangle$?

Solution: We have $(3+i)(3-i) = 3^2 + 1^2 = 10$, so $10 \in \langle 3+i \rangle$. Thus the order of coset $1 + \langle 3+i \rangle \in \mathbb{Z}[i]/\langle 3+i \rangle$ is a divisor of 10. Actually the order is precisely 10, e.g. 5 is not the order since $5 \notin \langle 3+i \rangle$ (indeed $\frac{5}{3+i} = \frac{3-i}{2}$ so 5 is not a multiple of $3+i$ in the ring $\mathbb{Z}[i]$). Also $a+bi = a-3b + b(3+i)$, so an arbitrary coset $a+bi + \langle 3+i \rangle \in \mathbb{Z}[i]/\langle 3+i \rangle$ contains a representative (e.g. $a-3b$) from $\mathbb{Z}$. In other words the abelian group $\mathbb{Z}[i]/\langle 3+i \rangle$ is generated by coset $1 + \langle 3+i \rangle$. Thus $\mathbb{Z}[i]/\langle 3+i \rangle$ is a cyclic group with generator of order 10.

Answer: the ring $\mathbb{Z}[i]/\langle 3+i \rangle$ has 10 elements.

Comment. In fact, the ring $\mathbb{Z}[i]/\langle 3+i \rangle$ is isomorphic to the ring $\mathbb{Z}_{10}$. Prove it!

Problem 5. Prove that $\langle 2+2i \rangle$ is not a prime ideal of $\mathbb{Z}[i]$. What is characteristic of $\mathbb{Z}[i]/\langle 2+2i \rangle$?

Solution: The elements $2, 1+i \in \mathbb{Z}[i]$ are not contained in $\langle 2+2i \rangle$ (e.g. $\frac{2}{2+2i} = \frac{1-i}{2} \notin \mathbb{Z}[i]$). However $2 \cdot (1+i) = 2+2i \in \langle 2+2i \rangle$, so the ideal $\langle 2+2i \rangle$ is not prime.

We have $(2+2i)(1-i) = 4$, so $4 \in \langle 2+2i \rangle$. Thus the order of coset $1 + \langle 2+2i \rangle \in \mathbb{Z}[i]/\langle 2+2i \rangle$ is a divisor of 4. Actually the order is precisely 4, e.g. 2 is not the order since $2 \notin \langle 2+2i \rangle$. Since the ring $\mathbb{Z}[i]/\langle 2+2i \rangle$ has a unity (which is the coset $1 + \langle 2+2i \rangle$), its characteristic is the order of the unity.

Answer: characteristic of $\mathbb{Z}[i]/\langle 2+2i \rangle$ is 4.

Comment. One can show that $\mathbb{Z}[i]/\langle 2+2i \rangle$ has 8 elements. However it is not isomorphic to $\mathbb{Z}_8$ (since the characteristic of $\mathbb{Z}_8$ is 8).

Problem 6. Prove that $\mathbb{Z}_2[x]/\langle x^2+x+1 \rangle$ is a field. How many elements are in this field?

Solution: $\mathbb{Z}_2$ is a field; so $\mathbb{Z}_2[x]/\langle p(x) \rangle$ is a field if and only if $p(x)$ is irreducible. The polynomial $p(x) = x^2+x+1 \in \mathbb{Z}_2[x]$ is irreducible: it is of degree 2, so if it is not irreducible it would have a zero in $\mathbb{Z}_2$; it does not since $p(0) = p(1) = 1$.

For any $p(x)$, $\mathbb{Z}_2[x]/\langle p(x) \rangle$ is a vector space over $\mathbb{Z}_2$ of dimension equal to the degree of $p(x)$. Thus $\mathbb{Z}_2[x]/\langle x^2+x+1 \rangle$ has dimension 2, so it is isomorphic to $(\mathbb{Z}_2)^2$ as a vector space over $\mathbb{Z}_2$. In particular it has $2^2 = 4$ elements.

Answer: $\mathbb{Z}_2[x]/\langle x^2+x+1 \rangle$ is a field with 4 elements.

Problem 7. Assume R is PID and let $N \subset R$ be an ideal. Prove that any ideal of R/N is principal. True/False: under the assumptions above, R/N is always PID.

Solution: Let $\phi : R \rightarrow R/N$ be the canonical homomorphism. Let $M \subset R/N$ be an ideal and let $\phi^{-1}(M) = \{x \in R \mid \phi(x) \in M\}$ be its pre-image; thus $\phi^{-1}(M)$ is an ideal of R . Since R is PID, the ideal $\phi^{-1}(M)$ is principal, so $\phi^{-1}(M) = \langle a \rangle$ for some element $a \in R$. We claim that $M = \langle \phi(a) \rangle$. Indeed $\phi(a) \in M$ since $a \in \phi^{-1}(M)$; hence $\langle \phi(a) \rangle \subset M$. Conversely if $m \in M$ then there exists $r \in R$ such that $\phi(r) = m$ (since ϕ is surjective). Hence $r \in \phi^{-1}(M) = \langle a \rangle$, so $r = as$ for some $s \in R$. Then $m = \phi(r) = \phi(as) = \phi(a)\phi(s) \in \langle \phi(a) \rangle$, so $M \subset \langle \phi(a) \rangle$. Since $\langle \phi(a) \rangle \subset M$ and $M \subset \langle \phi(a) \rangle$, we have $M = \langle \phi(a) \rangle$. Thus any ideal of R/N is principal.

The answer to True/False question is False: sometimes R/N is not integral domain (e.g. for $R = \mathbb{Z}$ and $N = \langle 4 \rangle$), so R/N is not PID in such cases.

Problem 8. Let $\phi : \mathbb{Z}[i] \rightarrow \mathbb{Z}_{13}$ be a unital ring homomorphism (such homomorphisms do exist). What are possible values of $\phi(i)$?

Solution: Since $i^2 = -1$, we get $\phi(i)^2 = \phi(-1) \equiv 12 \equiv 5^2 \pmod{13}$. Thus $(\phi(i) - 5)(\phi(i) + 5) = 0$. Since $\mathbb{Z}_{13}$ is integral domain this implies that either $\phi(i) = 5$ or $\phi(i) = -5 \equiv 8 \pmod{13}$.

Answer: $\phi(i) = 5$ or $\phi(i) = 8$.

Comment. In fact, there are precisely two unital homomorphisms $\mathbb{Z}[i] \rightarrow \mathbb{Z}_{13}$ sending i to 5 or 8. Show that!

Problem 9. Show that the rings $\mathbb{Z}[x]/\langle x^2 + 2x + 3 \rangle$ and $\mathbb{Z}[\sqrt{-2}]$ are isomorphic.

Solution: Consider evaluation homomorphism $ev : \mathbb{Z}[x] \rightarrow \mathbb{Z}[\sqrt{-2}]$ sending x to $-1 + \sqrt{-2}$. This homomorphism is surjective since $ev(a + b + bx) = a + b\sqrt{-2}$. The kernel $\text{Ker}(ev)$ consists of polynomials $p(x) \in \mathbb{Z}[x]$ such that $p(-1 + \sqrt{-2}) = 0$. This implies that $p(-1 - \sqrt{-2}) = 0$ since $-1 - \sqrt{-2}$ is complex conjugate of $-1 + \sqrt{-2}$. Thus $p(x)$ is divisible (as polynomial with real coefficients) by $(x - (-1 + \sqrt{-2}))(x - (-1 - \sqrt{-2})) = x^2 + 2x + 3$. Long division implies that $\frac{p(x)}{x^2 + 2x + 3}$ has integer coefficients (this also follows from Gauss' Lemma), that is $p(x) \in \langle x^2 + 2x + 3 \rangle$. Conversely, for any polynomial $p(x) \in \langle x^2 + 2x + 3 \rangle$ we have $p(-1 + \sqrt{-2}) = 0$. Thus $\text{Ker}(ev) = \langle x^2 + 2x + 3 \rangle$ and by the Homomorphism Theorem $\mathbb{Z}[x]/\langle x^2 + 2x + 3 \rangle \simeq \mathbb{Z}[\sqrt{-2}]$.

Problem 10. Show that the rings $\mathbb{Z}[\sqrt{-2}]$ and $\mathbb{Z}[i]$ are not isomorphic.

Solution: Assume that $\phi : \mathbb{Z}[\sqrt{-2}] \rightarrow \mathbb{Z}[i]$ is an isomorphism. Then $x = \phi(\sqrt{-2})$ satisfies $x^2 = \phi(\sqrt{-2})^2 = \phi((\sqrt{-2})^2) = \phi(-2) = -2$. However for $x = a + bi$, $a, b \in \mathbb{Z}$ we have

$$x^2 = (a + bi)^2 = a^2 - b^2 + 2abi,$$

and the equation $x^2 = -2$ is equivalent to system of equations

$$\begin{cases} a^2 - b^2 &= -2 \\ 2ab &= 0 \end{cases}$$

which clearly has no integer solutions (since the second equation implies that $a = 0$ or $b = 0$).